

BASE (NOSQL) Properties

Prepared By: Dr. D. P. Singh

Overview

- The BASE properties of a database management system are a set of principles that guide the design and operation of modern databases.
- The acronym BASE stands for Basically Available, Soft state, and Eventual consistency.
- BASE is often associated with systems such as Apache Cassandra, Amazon DynamoDB, and Apache CouchDB.

1. Basically Available (BA)

- The system remains available and responds to requests even when some nodes fail or communication problems occur.
- **Example**
- Consider an online shopping website with data replicated across multiple servers.
 - Server A ✓
 - Server B ✗ (failed)
 - Server C ✓
- If Server B crashes:
 - Users can still browse products.
 - The system continues serving requests from A and C.
 - Some data may be slightly outdated.
- **Key Point**
 - Availability is preferred over immediate consistency.

2. Soft State (S)

- The state of the system may change over time even without new user input because replicas synchronize asynchronously.
- **Example**
- A social media post receives a new "Like."
 - Server A = 101 likes
Server B = 100 likes
Server C = 100 likes
- A few seconds later, replication occurs:
 - Server A = 101 likes
Server B = 101 likes
Server C = 101 likes
- The system's state changes automatically as replicas exchange updates.
- **Key Point**
 - Data is not required to be immediately consistent at all nodes.

3. Eventual Consistency (E)

- If no new updates are made, all replicas will eventually converge to the same value.
- **Example**
- A user updates their profile picture.
 - Immediately after update:
 - Server A = New Photo
 - Server B = Old Photo
 - Server C = Old Photo
- After replication:
 - Server A = New Photo
 - Server B = New Photo
 - Server C = New Photo
- All servers eventually contain the same data.
- **Key Point**
- Temporary inconsistencies are acceptable as long as synchronization eventually occurs.

Real-World Example: Social Media Likes

- Suppose a post gets a new like.

- **Initially**

- Replica A = 500 likes
 - Replica B = 500 likes
 - Replica C = 500 likes

- **User clicks Like**

- Replica A = 501 likes
 - Replica B = 500 likes
 - Replica C = 500 likes

- Some users may see:

- 500 likes

- Others may see:

- 501 likes

- After synchronization:

- Replica A = 501
 - Replica B = 501
 - Replica C = 501

This demonstrates:

Basically Available → service remains operational.

Soft State → values change during replication.

Eventual Consistency → replicas eventually agree.

Advantages and Disadvantages

• Advantages of BASE

- ✓ High availability
- ✓ Horizontal scalability
- ✓ Better fault tolerance
- ✓ Suitable for globally distributed systems
- ✓ Handles massive traffic efficiently

Disadvantages of BASE

- ✗ Temporary stale data
- ✗ Different users may see different values briefly
- ✗ More complex application logic
- ✗ Difficult conflict resolution in some cases

BASE vs ACID

Feature	BASE	ACID
Consistency	Eventual	Strong
Availability	High	Lower during failures
Scalability	Excellent	More limited
Transactions	Relaxed	Strict
Suitable For	Large distributed systems	Financial and transactional systems
Example Use Cases	Social media, caching, analytics	Banking, payments, reservations

Thank You